PPP-T-42C
January 2, 1975
SUPERSEDING
Fed. Spec. PPP-T-42B
July 16, 1973, and
Int.Fed.Spec.UU-T-00106D(GSA-FSS)
April 22, 1970, and
Fed. Spec. UU-T-106C
July 27, 1959

FEDERAL SPECIFICATION

TAPE, PACKAGING/MASKING, PAPER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of paper tape to be used for general light-duty packaging and bundling, and for temporarily masking surfaces not to be covered by finishing materials (see 6.1).

1.2 Classification. The paper tape shall be of the following types as specified (see 6.2):

Type I - Crepe.
Type II - Flat.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-T-680 - Tape, Pressure-Sensitive Adhesive: Packaging and Packing Of.

Federal Standard:

Fed. Test Method Std. No. 147 - Tapes, Packaging, Adhesive and Gummed: Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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3.1 Construction. The paper backing shall be coated over the entire surface of one side with an adhesive that requires no moisture, heat, or other preparation prior to application. The adhesive shall cause the tape to adhere immediately and firmly to clean, dry surfaces and shall not transfer when the tape is unwound from the roll. The paper shall perform satisfactorily as a masking media when used with paint, varnish, or other finishing materials. The paper shall have sufficient strength to allow the tape to be unwound from the roll and to be removed from the masked surface after application of finishing materials without breakage or delamination. The finishing materials shall not penetrate the paper sufficiently to have a deletorious effect on the adhesive.

3.2 Rolls. The tape shall be uniformly and evenly wound on 60-yard rolls, adhesive side in, on cores of sufficient rigidity to prevent distortion of the rolls under normal conditions of transportation, storage, and use. The inside diameter of the core shall be 3 - 0 + 1/16 inches. No more than three splices will be permitted in any roll. All splices shall be such that they will not separate when the roll is unwound in test or application.

3.2.1 Width. The width of the tape shall be 1, 2, or 3 inches or any other commercially available widths specified in the contract or order (see 6.2). A tolerance of ± 1/32 inch shall be allowed on all widths.

3.2.2 Color. Unless otherwise specified in the contract or order (see 6.2), the color shall be natural.

3.3 Physical properties. The tape shall meet the requirements listed in table I when tested as described in Section 4.

TABLE I.

	Type I	Type II	Test
Adhesion, oz./in. (min.)			
Initial	25	25	4.4.2
Aged	20	20	4.4.3
To backing	8	--	4.4.4
Elongation, percent			
Min.	6	--	4.4.2
Max.	15	--	4.4.2
Tack (rolling ball), inches (max.)	6	--	4.4.5
Tensile, pounds/inch (min.)	16	25	4.4.2
Thickness, inches (max.)	0.009	0.009	4.4.2

3.4 Workmanship. Workmanship shall be first class throughout. The tape shall be free of defects which detract from its appearance or impair its serviceability. The adhesive shall be a smooth uniform coating, covering the entire area of one side of the tape. Edges shall be straight, true, and unbroken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 End item examination. Sampling and examination of the item shall be as specified in Section 7 of Fed. Test Method Std. No. 147.

4.3 End item testing. Sampling and examination for end item testing shall be as specified in Section 7 of Fed. Test Method Std. No. 147 at an Acceptable Quality Level (AQL) of 4 percent.

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4.4 Tests. Tests for acceptance of individual lots shall consist of tests for all requirements specified in Section 3.

4.4.1 Specimen. The specimen shall be selected and prepared as described in Method 4.1 of Fed. Test Method Std. No. 147.

4.4.2 Test methods. The following tests shall be conducted as described in Fed. Test Method Std. No. 147, except that one specimen per sample unit shall be tested:

Adhesion, initial	- Method 10.2
Tensile	- Method 30.2
Elongation	- Method 30.2
Accelerated aging	- Method 60.2
Thickness	- Method 36.2

4.4.3 Aged adhesion. Rolls of 10 yards minimum shall be aged as described in Method 60.2 of Fed. Test Method Std. No. 147 for 7 days. After aging, the tape shall be tested for adhesion as described in Method 10.2 of Fed. Test Method Std. No. 147.

4.4.4 Adhesion to backing. The adhesion to backing shall be conducted as described in Method 10.2 of Fed. Test Method Std. No. 147, except that the test panel shall be covered with a strip of the tape under test prior to application of the test specimen. Care should be taken when applying the tape to cover the test panel that the backside of the tape is not touched with the fingers. The tape covering the panel shall be at least as wide as the test specimen.

4.4.5 Tack (rolling).

4.4.5.1 Apparatus.

(a) Incline - A special incline shown in figure 1, is required.
(b) Steel ball - 1.0625 + 0.0100 inch diameter standard ball bearing is required. Weight shall be 80.0 + 1.0 grams.
(c) Working surface - A level, hard, smooth surface, such as a table top, is required.

4.4.5.2 Procedure. Arrange a tape specimen 15 inches long, adhesive side up, in line with the raceway of the incline, as shown in figure 2. The tape specimen shall be free of any wrinkles, creases, or splices. The end of the tape opposite the incline may be held to the table with a weight or with other tape. If sides of the tape specimen have a tendency to lift or curl, they shall be securely held to the table with tape. Prior to testing each lot of tape, thoroughly clean the raceway surface with N-heptane, methyl ethyl ketone, isopropyl alcohol, or methyl alcohol. Prior to each roll of the ball, thoroughly clean the ball with the same solvent. Wipe with clean, dry surgical gauze to remove any remaining residue. After cleaning, do not touch the raceway or ball with fingers. Using clean, dry tongs, place the ball on the upper side of the release. Release the ball and allow it to roll to a stop on the adhesive. Measure the distance from the center of contact between the ball and adhesive to the near end of the incline as shown in figure 2. Repeat the procedure for each specimen.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be in accordance with the requirements of PPP-T-680. Unless otherwise specified (see 6.2), bulk packaging shall be used. The level of packaging and packing shall be A, B, or C, as specified in the contract or order (see 6.2).

5.2 Marking. Containers of tape shall be marked in accordance with the requirements of PPP-T-680. When specified (see 6.2), the core of each roll of tape shall be marked on the inside or on the edge with numerals or letters indicating the month and year of manufacture and specification number. In addition, the manufacturer's name and designation of the product shall be marked on the core.

6. NOTES

6.1 Intended uses.

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6.1.1 Type I tape is primarily intended for light-duty bundling, holding, and packaging applications such as bundling small parts to be overpacked, holding small parts to larger assemblies, and for temporary closing of chipboard boxes and fiberboard boxes that are not to be shipped between bases. This tape is not intended to be a substitute for PPP-T-76, carton sealing tape. Type I tape is also for use in temporarily masking surfaces from being covered by paint, varnish, or other finishing materials. For masking applications, type I is for use on contoured or rounded surfaces.

6.1.2 Type II tape is only for use in temporarily masking surfaces from being covered by paint, varnish, or other finishing materials. Type II is for use where straight line color separation is desired.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of tape required (see 1.2).
(c) Width of tape required (see 3.2.1).
(d) Color, if other than natural (see 3.2.2).
(e) Whether individual or bulk packaging is to be used (see 5.1).
(f) Selection of applicable levels of packaging and packing (see 5.1).
(g) Marking required (see 5.2).

6.3 Storage. The tape should be stored in the original container in a cool location. It should not be stored in close proximity to steam pipes, radiators, or other sources of heat, or in a location where it may be exposed to direct rays of the sun.

6.4 Relationship to superseded types. PPP-T-42B tape is the same as type I of PPP-T-42C. Type I of UU-T-00106D is similar to PPP-T-42B and type I of PPP-T-42C, except that PPP-T-42B and type I of PPP-T-42C may be used for light-duty packaging and bundling. This was not the case with type I of UU-T-00106D. Type II of UU-T-00106D is the same as type II of PPP-T-42C. UU-T-00106D is similar to UU-T-106C, except the requirements were improved.

MILITARY INTERESTS:
CUSTODIANS:
Army - GL
Navy - SA
Air Force - 69

Review Activities:
Army - SM, MD

CIVIL AGENCY COORDINATING ACTIVITIES:
HEW-FDA, NIH
NASA-JFK
DOT-ACO
TVA

PREPARING ACTIVITY:
GSA-FSS

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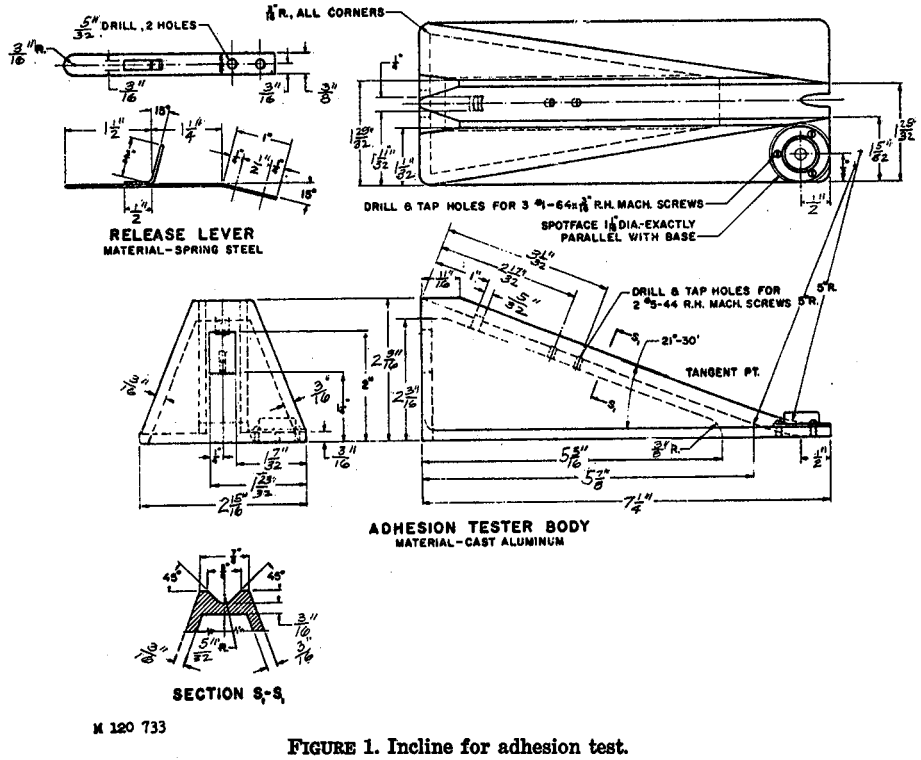


FIGURE 1. Incline for adhesion test.

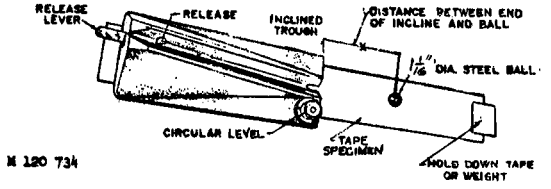


FIGURE 2. Adhesion test apparatus and specimen showing distance of roll that is measured.

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